

README

Tamil AI Learning Assistant

Overview

Tamil AI Learning Assistant is an AI-powered application designed to help users learn, write, and communicate in Tamil efficiently. It focuses on converting Tanglish (Tamil written in English letters) into grammatically correct Tamil and enhancing language quality using AI.

This system combines modern mobile development, backend engineering, and local AI inference to deliver a complete learning solution.

Problem Statement

Typing correct Tamil is difficult for many users, especially those accustomed to Tanglish. This creates barriers in:

- Learning Tamil grammar
- Writing correct sentences
- Communicating effectively

Solution

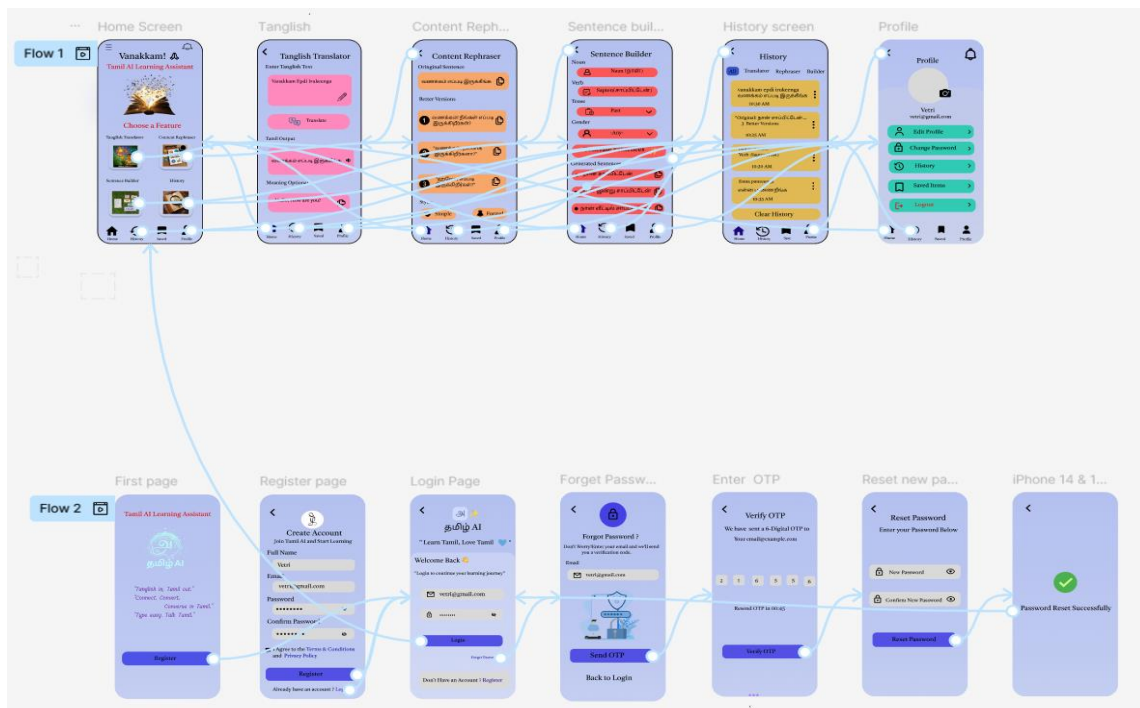
This application solves the problem using AI by:

- Converting Tanglish → Proper Tamil
- Improving grammar and sentence structure
- Rephrasing Tamil content
- Generating Tamil sentences based on inputs

Core Features

- Tanglish → Tamil Translator
- Tamil Rephraser (3 versions output)
- Sentence Generator (based on grammar input)
- User Authentication (JWT based)
- Mobile-friendly UI (Flutter)
- Fast AI inference using Ollama

System Architecture



Tech Stack

Frontend

- Flutter
- Dart

Backend

- Spring Boot
- REST APIs
- JWT Authentication
- Maven

AI Engine

- Python
- FastAPI
- Ollama (Gemma 4:e2b)

Database

- MySQL

Workflow

1. User enters Tanglish input in mobile app
2. Flutter sends request to Spring Boot backend
3. Backend forwards request to FastAPI AI service
4. AI model processes input (Ollama)
5. Response returned → Backend → Flutter
6. Output displayed to user

Key Modules

1. Translation

- Converts Tanglish → Tamil
- Fixes grammar and slang

2. Rephrase

- Generates 3 improved Tamil versions
- Returns structured JSON

3. Sentence Generator

- Generates Tamil sentences based on:
 - Noun
 - Verb
 - Tense

Authentication

- JWT-based login system
- Tokens stored locally in app
- Persistent session (no re-login required)

Strengths

- End-to-end full-stack architecture
- AI integration with local model
- Real-time Tamil processing
- Clean UI and user flow

Limitations

- Requires backend + AI service running
- AI response time depends on system performance (CPU-based inference)

Future Enhancements

- Voice to Tamil conversion
- Offline AI support
- Pronunciation assistant
- Cloud deployment

Notes

Ensure all services are running before launching the app:

- Spring Boot Backend
- FastAPI AI Service
- Ollama Model